

calc — sheet: inputs

Parameter	Value	Source
Positive electrode thickness [m]	7.56e-05	parameter set dump
Negative electrode thickness [m]	0.0001022	bridge/params_final.json
Separator thickness [m]	1.2e-05	parameter set dump
Positive CC thickness [m]	1.6e-05	parameter set dump
Negative CC thickness [m]	1.2e-05	parameter set dump
Positive electrode porosity	0.335	parameter set dump
Negative electrode porosity	0.25	parameter set dump
Separator porosity	0.47	parameter set dump
Positive active material volume fraction	0.665	parameter set dump
Negative active material volume fraction	0.75	parameter set dump
Positive particle radius [m]	5.22e-06	parameter set dump
Negative particle radius [m]	3.5e-06	bridge/params_final.json
Positive electrode density [kg.m-3]	4400	parameter set dump
Negative electrode density [kg.m-3]	1657	parameter set dump
Maximum concentration in positive electrode [mol.m-3]	63104	parameter set dump (Chen2020; LNMO.json key inert - recorded)
Maximum concentration in negative electrode [mol.m-3]	33133	parameter set dump
Electrode height [m]	0.065	parameter set dump
Electrode width [m]	1.58	parameter set dump
Electrode area [m2]	0.1027	cell/r8_d6_energy_dfn.json:area_m2
Lower / Upper voltage cut-off [V]	2.5 / 4.7	parameter set dump
Cation transference number	0.2594	parameter set dump
Electrolyte diffusivity [m2.s-1]	6e-10	bridge/params_final.json
EC initial concentration in electrolyte [mol.m-3]	2270.5	bridge/params_final.json
EC diffusivity [m2.s-1]	1e-18	bridge/params_final.json
Total heat transfer coefficient [W.m-2.K-1]	400	bridge/params_final.json
Faraday constant [C.mol-1]	96485	physical constant

calc — sheet: capacity_energy

Quantity	Value	Formula	Source
1C discharge capacity [Ah]	6.1976	time integral of I at 1C to 2.5 V	cell/r8_d6_1c_dfn.json:capacity_ah
Discharge energy [Wh]	25.4059	integral(V x I_1C) dt	cell/r8_d6_energy_dfn.json:energy_wh

Quantity	Value	Formula	Source
Midpoint voltage [V]	4.1145	V at t=len(t)/2 (DFN uniform grid, ~70% depth)	cell/r8_d6_energy_dfn.json:midpoint_voltage_v
Cell volume [m3]	2.2368e-05	sum(layer thickness) x area	cell/r8_d6_energy_dfn.json:volume_m3
DCR [Ohm]	0.00638	midpoint DCR (calc-energy)	cell/r8_d6_energy_dfn.json:dcr_ohm

calc — sheet: energy_density

Quantity	Value	Formula	Source
Volumetric ED [Wh/L]	1135.8	energy_wh / volume_m3 / 1000	cell/r8_d6_energy_dfn.json:energy_density_wh_l
Gravimetric ED [Wh/kg]	493.3	energy_wh / mass_kg	cell/r8_d6_energy_dfn.json:energy_density_wh_kg
Mass [kg]	0.0515	sum(layer_kg_m2 x area_m2)	cell/r8_d6_energy_dfn.json:mass_kg
Criterion check	1135.8 >= 950 -> PASS (+19.6%)		entry 0 criteria.stage2

calc — sheet: np_mass

Quantity	Value	Formula	Source
Positive areal capacity [Ah/m2]	85.03	$F \times c_{\text{max_pos}} \times \text{eps_am_pos} \times L_{\text{pos}} / 3600 = 96485 \times 63104 \times 0.665 \times 75.6\text{e-}6 / 3600$	parameter set dump
Negative areal capacity [Ah/m2]	68.07	$F \times c_{\text{max_neg}} \times \text{eps_am_neg} \times L_{\text{neg}} / 3600 = 96485 \times 33133 \times 0.75 \times 102.2\text{e-}6 / 3600$	parameter set dump
N/P ratio	0.8	68.07 / 85.03 (capacity ratio; supersedes round shorthand 'N/P 1.2' = thickness-ratio mislabel - recorded)	computed from parameter set
Layer mass Positive electrode coating (LNMO) [kg]	0.022718	layer_kg_m2 x area_m2	cell/r8_d6_energy_dfn.json
Layer mass Negative electrode coating (graphite) [kg]	0.013044	layer_kg_m2 x area_m2	cell/r8_d6_energy_dfn.json
Layer mass Positive current collector (Al) [kg]	0.004437	layer_kg_m2 x area_m2	cell/r8_d6_energy_dfn.json
Layer mass Negative current collector (Cu) [kg]	0.011042	layer_kg_m2 x area_m2	cell/r8_d6_energy_dfn.json
Layer mass Separator [kg]	0.000259	layer_kg_m2 x area_m2	cell/r8_d6_energy_dfn.json
Total mass [kg]	0.0515	sum of layers	cell/r8_d6_energy_dfn.json:mass_kg

calc — sheet: process

Parameter	Value	Formula	Source
Positive areal density [g/m2]	221.2	$75.6\text{e-}6 \times 0.665 \times 4400 \times 1000$	parameter set dump
Negative areal density [g/m2]	127	$102.2\text{e-}6 \times 0.75 \times 1657 \times 1000$	parameter set dump
Positive compaction density [g/cm3]	2.93	$4400 \times 0.665 / 1000$	parameter set dump
Negative compaction density [g/cm3]	1.24	$1657 \times 0.75 / 1000$	parameter set dump

Parameter	Value	Formula	Source
Pore volume [cm3]	5.804	$\text{sum}(\text{thickness} \times \text{porosity} \times \text{area}) \times 1\text{e}6$	parameter set dump
Electrolyte fill [g]	6.27	$5.804 \times 1.2 \times 0.9$ (density = literature 1.2 g/cm3)	design estimate
Formation	0.1C CC to 4.7 V, 25 C, 2 cycles		design-recommended value; production tuning required